Seat Selector

The teacher and teaching assistants of ITU in general have a great time teaching the BootIT Course at ITU. However, they would like you to make it even better! Every year the teachers try to get the students to keep every third row of seats free in the auditorium, in an attempt to optimize the help process. Nevertheless, it is sometimes hard to remember to tell people as well as seeing which row is the third. Therefore, the teachers have hired you! They would like you to create a Java program, which can help the poor teachers in knowing what rows should be kept free. Thus, making their life even better.

Functional Requirements

The teacher and teaching assistants have some specific requirements for the system to max out the help process.

1. The teachers often use different rooms, so there is a different number of rows. Therefore, the program needs to be dynamic and handle different rooms. The program therefore has to start with asking about how many rows there is a room.
   (Green: terminal output, Blue: User Input)

```
> How many rows are there in the room

> 10

> Coolio! I've computed which rows that must be free now!
```

2. The program needs to be able to tell the teachers if a row should be kept free. Below an example is shown
   (Green: terminal output, Blue: User Input)

```
> Which row would the student like to sit in?

> 3

> That's an invalid row! That row must be free! Try another
one…

> 4

> That's a brilliant choice. Please take a seat an enjoy the
smooth voice of Niek
```

3. The program should also be able handle unreasonable sitting request.

```
> Which row would the student like to sit in?

> 5000

> Sorry, there is only 10 rows?! Please try another.

> -500

> Negative numbers?! Creative… We are teaching IT here, not how
to move through matter. Please pick another seat.
> 0

> Really? The floor? Well maybe try to select another…
```

4. The program must be able to stop when the lecture starts. When doing so the teachers would potentially like to know how many free rows there are. Aiding them in knowing how they should locate them self in the rooms.

```
> Which row would the student like to sit in?

> STOP

> Cool, everyone must be seated then! Please enjoy the lecture. Do
you want to know how many free rows there are (y/n)


/* If yes */
> y

> According to my calculations, there should be a total of 3 rows
there should be free.

/* If No */
> n

> Okay… Then count it yourself…
```

Cool Additional Features

The requirements described in the previous chapter are all the teachers' needs and all that they are willing to pay for. Nevertheless, Jens is always trying to see if he can get a bit extra on top… Consequently, he has sent an additional mail with different request, which he thinks could be cool…

What up, cool developers!

So I've heard about the requirements of the system! Sounds pretty cool! However, I have some ideas that will make the product even cooler! So, I've made a short list of potential addons.

- **I would like the system to tell me how many students have sat down during the day!**
- **I've started to do some carpeting at home. Sometimes I build extra rows for auditoriums. I would like to know if I add a row to the total number of rows, whether it should be free or not? - You know so I can save some money on materials (Then it doesn't have to be so sturdy)**
- **I was wondering if you could make the output messages of the system even cooler?**

I hope you'll consider my requests!

Best regards
Jens

Remember the ideas of Jens are not part of the original requirements. It's up to you if you would like to implement any of his ideas.

Hints

The teachers have come with some hints for the creation of this program.

- The Scanner might be a nice tool to get some input
- Perhaps you can use the Modulo (%) operator for some of it.
- The solution can be done with loops. But also without (difficult).